Lesson Notes

MODULE 4 | LESSON 2

LEVERAGE AND NON-LINEARITY

Reading Time	60 minutes
Prior Knowledge	Option payoffs, Calls, Puts, Return Calculations
Keywords	Leverage, To Lever, Timing, Volatility Trader, Straddle, Non-linearity

In the previous lesson, we outlined the basics of options: calls and puts, premiums paid upfront, potential payoffs if the option expires in the money, or nothing if it expires out of the money. Options depend on their underlying stock, the strike level, the time to expiration, the risk-free rate, the dividend yield, and the stock's volatility. In this lesson, we dive deeper into the leverage and nonlinearity features of options.

1. Leverage

We mentioned leverage several times in this and in previous lessons. In Lesson 2, we are going to discuss options as an investment choice, and we'll see how leverage comes into play when working with options.

The first thing we'll discuss is the **leverage** that's built into options. A possible definition of leverage is that we borrow capital to invest more deeply in an investment than merely by investing with our cash on hand. The return on capital is expected to exceed the rate of borrowing, so leverage is a double bet: that the investment itself will have a positive return, and that the rate of return exceeds the cost of borrowing so that it will have a net positive return on borrowed funds as well.

The beauty of options is indeed that they are a leveraged investment because they **leverage** the returns. How? Not by borrowing funds. Indeed, **an option can only be paid for in full**. Instead, options use leverage through a conversion factor greater than 1. The conversion factor is the number of shares of stock that can be traded for one option. For stock options, the conversion factor is 100. One long equity option entitles the holder to trade 100 shares of the underlying stock. Buying one call option would grant the buyer the right (and not the obligation) to buy 100 shares of stock at the strike level. Note that this option costs less than buying 100 shares. So for a lower initial investment than buying stocks, or even buying stock on margin, the equity option offers a leveraged approach to investing.

Let's compare the approaches of buying stock with cash, buying stock with financing, and buying options. Suppose we are bullish on a stock on a company. Let's examine each of these ways that we could invest in that company. Let's imagine that the company is expected to make a return of 10%.

2. Buying Stock Using Cash

The first thing we could do is buy the stock using cash. Suppose that the stock price is $500 per share. We want to trade 100 shares. We need $500/share * 100 shares = $50,000. Recall our expected return is 10%. So our $50,000 would increase 10% or $5,000. Our return is $5,000 on a $50,000 investment or 10%. Buying stocks for cash is unleveraged. The returns are in no way magnified because no debt was incurred to borrow funds and no derivatives were used to increase the exposure size. This is the most expensive way to make $5,000 because we need $50,000 of capital upfront to do so.

3. Buying Stock Using Financing

Consider a second example: we're buying stocks using financing. This is indeed what many hedge funds do. For example, let's say that you bought $100,000 worth of stock. Now, $50,000 of that amount is your actual cash on hand. The other $50,000 is borrowed. Of course, you will have to pay a financing fee for the cost of those funds. However, suppose the stock indeed increases 10% overnight. Assuming the financing costs are negligible for the overnight rate, it appears we make $10,000 on $100,000, which is 10%. However, remember that only $50,000 was the initial investment. Indeed, once you return the other $50,000 that was borrowed, we can see that the return is actually $10,000 on an initial $50,000 investment or 20%. In other words, when we leveraged 2:1, our return was doubled from 10% to 20%.

$$\textbf{leveraged return} = \frac{\textbf{amount invested} + \textbf{amount borrowed}}{\textbf{amount invested}} \times \textbf{raw return}$$

In our example, we have:

$$\textbf{leverage return} = \frac{\mathbf{50,000+50,000}}{\mathbf{50,000}} \times \mathbf{10}$$

The beauty of leverage is that it can take small returns and make them larger! It can also help you achieve high returns more quickly. Leverage can help investments earn higher returns.

Suppose we only had $25,000 cash. We could borrow another $25,000, making our total investment $50,000. Then, we earn 10% on the total, which is $5,000. Since we only put in $25,000, we actually earn 20%. Leveraging 2 to 1 means we double our return, regardless of our starting position. The point here is that we needed only $25,000, and not $50,000, to earn $5,000.

In the examples involving financing, recall that we will have to pay some interest for borrowing funds. That's why we hypothesize the returns happen quickly, so that we can ignore these. In reality, though, if the investment is strong, the return should exceed the borrowing rates.

Let's consider another optimistic example. Suppose you could put down only 5% of the cost, say $5,000 and borrow 95% or $95,000. What would our leveraged return be if the raw return were 10%?

$$\textbf{leveraged return} = \frac{\mathbf{5,000+95,000}}{\mathbf{5,000}} \times \mathbf{10}$$

Leverage has turned a simple 10% investment into one that triples the money. We start with $5,000 and end up with $15,000 (our original $5,000 and a gain of $10,000). Before getting too excited, note that there are restrictions in different stock exchanges for the amount that can be borrowed. To prevent excessive leverage, there is regulation that caps the leverage for stocks for most investors at 50%.

Our examples have been optimistic. Let's switch to pessimistic examples. Suppose we were leveraged 2 to 1, or 20 to 1, and had a 10% price drop:

$$\textbf{leveraged return} = \frac{\mathbf{50,000+50,000}}{\mathbf{50,000}} \times \mathbf{-10}$$

Now instead of making 20%, we lose 20%. Leverage causes us to have extreme losses. Like volatility, leverage is indifferent to direction. Two to one leverage can just as easily give a big return or a very small return. The problem with extremely negative returns is that we may not be able to withstand the loss in the portfolio.

$$\text{leveraged return} = \frac{5,000 + 95,000}{5,000} \times -10$$

Indeed, with leverage, it is possible to lose your entire investment. It is even possible to lose more than your investment if you are highly leveraged. In this example, we started with \$5,000 and lost an additional \$10,000, for a total loss of \$15,000. This is -200%. This is one of the reasons that regulation exists: to limit the amount of leverage and avoid catastrophic losses.

To recap: leverage multiplies a raw return by the ratio of the total investment to your original investment. Leverage is indifferent to whether the return is positive or negative; it is a mere multiplier. Consequently, leverage can make a lot of money quickly, or it can also lose a lot of money quickly. Indeed, there are leveraged investments that cause losses people cannot sustain and can wipe out an entire investment portfolio. Being overleveraged is a risk. Prudent credit lenders will restrict the amount of leverage based on portfolio size, experience, and the volatility of the exchange.

4. Buying Options

Enter options. Options can only be bought fully with cash. That is, the option itself cannot be purchased using borrowed funds. This is because options already have leverage in them through a conversion factor rather than through borrowed funds. Recall that the conversion factor is 100: one option controls 100 shares of stock. If the stock price went up a dollar, in theory, the option should be worth about \$100 more because each of those shares that it controls increases by one. It's not as simple as that because the pricing is a complex combination of stock price and the other five factors discussed in the previous lesson. But the cost is going up a lot more than \$1 to be sure. Unlike the stock examples, there is not a hard and fast number like doubling or tripling on a raw return. However, it is possible for options to increase 100% or more. In other words, options have even more leverage than buying stocks on margin does. The leverage is *inherent* to the option.

5. Stocks or Options?

So clearly if you were bullish on a stock, you could decide to use all the funds to buy an option rather than buying the underlying or buying extra underlying with borrowed funds. What are the differences among these strategies?

Let's consider **timing**. Suppose we continued to believe the stock was going to increase 10%, but now we put a time horizon on that: within 1 month. So our three strategies would be:

1. Buy the stock with cash
2. Buy the stock with 50% cash and 50% borrowed funds
3. Buy a call option that expires within the month at a suitable strike

Now, suppose we get the time horizon wrong. It took 1 month and 2 days. The stocks make the same returns as in the previous example, but the option expires worthless. The stocks have the same returns because the underlying has no time constraint on earning a return; there is no expiration date. For the unleveraged stock investment, you're simply making the unleveraged return. For the leveraged stock, you would earn the leveraged return but have a slightly higher cost due to borrowing funds for a little more than a month. The double-digit return of the leveraged portfolio should more than compensate the investor for the extra interest on those borrowed funds. For the option however your return is -100%! That's right: it is possible to lose all your money by purchasing an option that then expires out of the money.

Options expire. If they do expire out of the money, they expire worthless. You had to pay a premium for that option; you don't get that premium back. The cost of that premium is effectively your loss. It's a -100% return, meaning a 100% loss. For the unsuspecting option trader, buying options can be daunting, as there are dozens of options to trade. Consider options on the company Meta, formerly known as Facebook. You could buy calls or you could buy puts. Suppose the stock price is currently trading at $200 per share. You could buy options with strikes at $2.50 intervals above and below the $200 price. In other words, there are calls and puts with strikes of $200, $197.50, $195, $192.50, … as well as $202.50, $205, $207.50, $210, etc. There are over twenty different strikes for which there are market makers offering to buy or sell options. Similarly, you could buy calls that expire this month, or one month from now, two months from now, three months from now, and so on. When you combine the numbers for types (calls and puts), strikes, and expiration dates, there could easily be over 100 options for a given stock. Which one do you choose?

In order to make money trading options, you have to get three things correct:

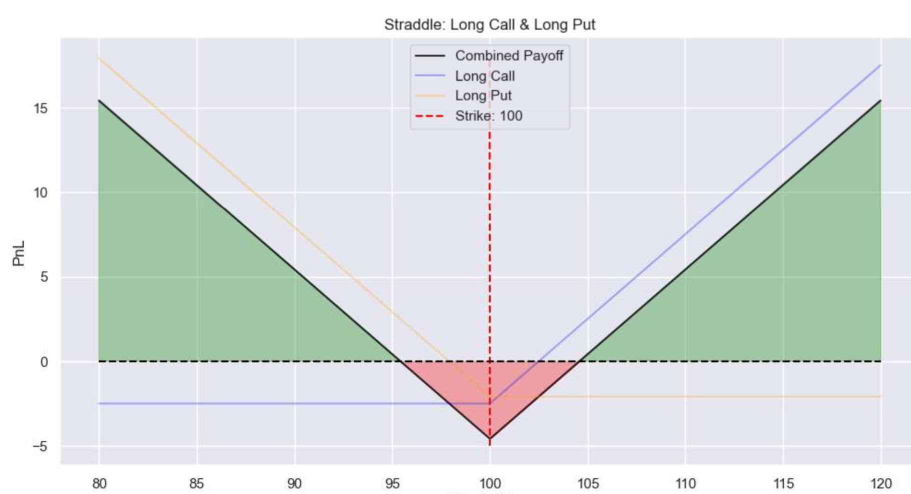
- Direction: Choose calls if the price should go up or puts if the price should go down.
- Timing: If you believe that it's going to take one month for your stock to go into the money above the strike, then make sure that you pick the right amount of time for that. In other words, make sure that there is enough time for the stock price to move into the money. Otherwise, the stock moving too slowly and not exceeding the strike will make the option expire worthless.
- Strike: Even if the stock price moved up, it may not move up far enough to be above the strike. For example, suppose you bought a one-month option when the stock price was $200 and your strike was $210. Suppose the final stock price was $209. You could have made money buying options with the strike of $205.00 or a strike of $207.50, but the strike of $210 produced a hurdle that the stock price did not overcome by its expiration.

Unfortunately, option traders don't get partial credit: getting only two out of three items (direction, timing, and strike) right means the option will expire out of the money.

6. Direction of Stock or Volatility

There is one way to avoid picking a direction. Suppose you are better at determining **volatility** than you are at estimating the direction the stock is headed. Instead of getting direction by choosing a call or a put, you can simply try to estimate the direction of volatility. In doing so, you can become a **volatility trader**. Suppose you were to buy a call and a put at the same strike. Here is what the payoff diagram looks like:

Figure 1: Straddle P&L



Now, what you are betting is the cost of two premiums for the stock's price to move a sizeable amount in either direction. Your total cost is the premium of a call and the premium of a put, both at the same strike. Since the cost is two premiums, the underlying stock has to move a considerable distance before there is a net profit. If the stock price moves off the strike level, then one of the options is in the money. However, this doesn’t mean that a payout amount is going to cover the cost of the two premiums. One of the two options will be exercised, but it may be insufficient to cover the payoff. A volatility trader is picking a direction of volatility rather than direction of the stock. The volatility trader who opts for this strategy, known as a straddle, believes volatility will increase. They are long volatility. If volatility does increase, they will make some money; it if increases enough, then they should make a profit.

When you’re long volatility, you hope for a big price move in either direction. If the price were to go down significantly, then the put is in the money and you can collect on that. Indeed, you cannot be in the money for both options at the same time. Conversely, if the stock price increases, then the call option is in the money.

If you believed the opposite, that volatility would drop, you could simply take the opposing side.

Suppose on the other hand you believe that volatility is going to drop. Then, you can sell the options. Selling the options would mean that you sell a call and you sell a put at the same strike. To be profitable, the stock price should not change too much. This position is short volatility. The maximum you can collect is two premiums. That's all the money you can make. However, if the stock price moves far enough in either direction, it's possible that you could lose money if one of the options is deep in the money. Note that in prior examples we're treating options as if we bought them, but in this example, we sold a pair of options, and that is where you have to be careful because you have to do something called hedging.

7. Option Payoff: Nonlinear

The option payoff--whether for calls or for puts--looks like hockey sticks. This is what is known as a **nonlinear** payoff. This is unlike anything we have seen. Bonds have a convex relationship to interest rates. Stocks and futures have linear payoffs. But options are different from everything else.

Let's examine the payoff of a call option. On the left side, we have a flat line. This flat line goes from 0 to K. (Since stock prices are nonnegative, we set the lower limit of the x-axis to 0.) This means that when the stock price is less than the strike level, the payoff for that call option is zero.

The next part of the payoff is where it slopes upward. This goes from K to infinity. Whenever the stock price exceeds the strike level, the option is in the money. When the stock price equals the strike level, the option is at the money (ATM). Even there, the payoff is zero, so the only positive payoff is when the stock price exceeds the strike level.

Each one of these segments is linear, but together, they combine to make the overall option payoff nonlinear. This is the key. Options have nonlinear payoffs, meaning that there is a "line in the sand"--the strike level--that is either crossed or not when the option expires.

As you remember, the payoff of the option only applies at the expiration. However, it would be helpful to know what the payoff would be during the lifetime of the option. Let's call it the intrinsic value. The intrinsic value is the option's payoff as if it could be exercised now. For a call option, the option's intrinsic value is zero when the stock price is less than the strike.

So here are three ways to say the same thing: an option with no intrinsic value is out of the money and would expire worthless if its expiration time was immediate.

As we have demonstrated, options have this nonlinear payoff. Where else do we see a nonlinear payoff? For this, let's turn next to the housing market.

8. Conclusion

In this lesson, we addressed the leverage and nonlinearity of options. In the next lesson, we will examine some remarkable similarities between options and mortgaged homes.